

Not Every Concept Is a Direction: Which Lexicalized Concepts Can Be Steered in LLMs? *

Anonymous ACL submission

Abstract

Steering vectors are widely used to control and interpret LLMs, yet benchmarks show that some concepts steer reliably while others barely respond — a variance that is reported but rarely explained. In this research we investigate which kinds of concepts are steerable in Llama-3.1-8B and Gemma-2-9B. We use the RFM method to extract steering vectors (or directions) for approximately 600 concepts drawn from 24 WordNet supersense categories, and validate which of them successfully steer the model. We then build a regression model explaining steerability from concept category, concreteness, polysemy, and word frequency, as well as the share of the top eigenvector during extraction (λ_1 /trace). The results help distinguish concepts that behave as single directions — “primitive” for the model — from those that are composite or context-dependent, and offer practitioners an extraction-time indication of when single-vector steering can be trusted.

- TODO: the prose paragraph above pre-dates the final results — update: add age-of-acquisition to the predictor list, replace “ λ_1 /trace” with “spectrum concentration (participation ratio)”, and consider adding one sentence each on the depth-window finding and the cross-model replication.
- Headline: 41.5% of the 600 concepts steer on Llama (249/600, V1UV6) and 37.3% on Gemma (224/600) under the v4 instrument, and the psycholinguistic laws replicate across BOTH models with all four predictors significant: frequency +, concreteness –, age-of-acquisition –,

polysemy –, Spectrum concentration (PR) predicts steerability within both models, at model-specific depths. Frequent, early-acquired, monosemous, abstract concepts steer best.

- Candidate sentence (failure mode): when steering fails, it fails informatively — delivery of the category/frame blend, a diffuse discriminative spectrum, or injection outside the concept’s depth window; not silence. (1 sentence)
- Candidate sentence (instrument): measured steerability is instrument-relative — one changed frame sentence doubles the rate on both models — but the laws survive instrument and model change. (1 sentence)
- Candidate sentence (depth): concepts are injectable in depth windows ordered by abstractness; one well-chosen layer nearly matches all-layer steering. (1 sentence)

1 Introduction (1.0p)

- Hook: steering/RePE papers throw heterogeneous concept lists at models; some concepts steer at 90%, others at chance. The variance is treated as noise; we argue it is signal. (1 para)
- The implicit assumption of top-1 eigenvector / PCA-1 / difference-of-means steering: the concept is representable as one direction, injectable with one coefficient, at whatever layer(s) the method uses. (half para)
- Existing evidence that not all features are one-dimensional (Engels et al.) — but no one has asked which concepts behave as single directions, or whether this aligns with human semantic structure. (half para)
- Our question: is steerability systematic and predictable from the linguistic properties of

*Working title. Alternative titles that fit the v4-era results: “Which Lexicalized Concepts Can Be Steered in LLMs — and at What Depth?”; the jul3 alternative (“Concreteness and Polysemy Predict...”) remains unusable (concreteness is negative; frequency positive).

the word — and what does a steering failure actually consist of? (half para)

- Contributions (short lead-in + three numbered items, 2–3 sentences each): (1 para)
 - C1: a systematic steerability study — ~ 600 lexicalized concepts across 24 WordNet supersenses, annotated with psycholinguistic norms, steered under a controlled protocol on two models; per-concept results and code released.
 - C2: a regression explaining steerability from frequency, concreteness, age-of-acquisition, and polysemy (all significant, signs replicating across Llama and Gemma), with extraction-internal spectral concentration (participation ratio / λ_1 -share) as a zero-cost predictor that beats the lexical block.
 - C3: a mechanistic account of what failure looks like — parent-dominance of extracted directions, coherence $\neq$ identity, injection outside the concept’s depth window — probed causally (within-category subtraction; single-layer depth sweep).

2 Background and Related Work (1.0p)

- Linear representation hypothesis and steering. Park et al.; RepE (Zou et al.); CAA (Panickssery/Rimsky et al.); ActAdd (Turner et al.); ITI (Li et al.); the RFM/universal-steering benchmark we build on (Beaglehole et al. 2026) — kernel ridge + AGOP metric learning, top eigenvector as steering vector. Ours: we adopt this family’s method (RFM) unchanged, but instead of proposing a steering technique we ask which concepts the technique works for and why. [NEW jul10] A second relation emerged: this literature’s success rates are instrument-relative — changing one sentence of the extraction contrast doubles the steerable share (19.8% $\rightarrow$ 41.5%) on the same method, model, and judge; benchmark comparisons that vary methods while hand-tuning per-class prompts inherit this confound (our per-class frame grid quantifies it). (1 para)
- Multi-dimensional features. Engels et al. (not all features are one-dimensional; cir-

cular features); feature splitting in SAEs; conceptors / subspace steering. Ours: same premise (one direction may not suffice), but we test it at the level of a stratified concept inventory and connect it to word properties rather than to individual features. [NEW jul10] Result: how concentrated a concept’s discriminative spectrum is (participation ratio / λ_1 -share) predicts steering success in both models — “effectively one-dimensional” is a measurable, graded property, and polysemy appears to act on steerability through it. (half para)

- [NEW jul6] Hierarchical concept geometry — must cite. Park, Choe, Jiang & Veitch (ICLR 2025, arXiv:2406.01506): ~ 900 WordNet concepts on Llama-3 and Gemma; categorical concepts are simplices and hierarchically related concepts occupy orthogonal subspaces. The representational-geometry twin of this study (same inventory source, same model families). Ours: they characterize how the hierarchy is encoded; we test what the hierarchy predicts about intervention — and our parent-dominance finding (extraction returns the shared category axis) is the steering-side counterpart of their parent/child orthogonal decomposition. [NEW jul10] R4 (averaging variant) has now run: subtracting the shared category axis does not rescue failed concepts (9 gained / 14 lost of 150) — consistent with their orthogonality result read causally: the child-specific residual exists but is rarely independently injectable; the shared axis also carries the steering drive. (half para)
- Predicting steerability. LAP / logit-lens accessibility (Billa 2026); “When is your LLM steerable?”; (un)reliability of steering vectors. Ours: these predict from engineering signals; we predict from the linguistic structure of the concept, and additionally explain what the failure mode consists of. [NEW jul10] Our lexical laws (frequency +, concreteness –, AoA –, polysemy –) replicate across two model families and two extraction instruments — portable, human-interpretable predictors that engineering-signal approaches cannot supply; conversely we contribute a cau-

tion their framing lacks: geometric predictors are only interpretable relative to the contrast design (our λ_1 -share flipped from null to significant when the frame changed). (half para)

- [NEW jul6] Add cites here: (a) “Understanding (Un)Reliability of Steering Vectors in Language Models” (arXiv:2505.22637) — directional agreement between per-example activation differences and the steering vector predicts steerability; (b) its 2026 follow-up on geometric predictors of steering unreliability (arXiv:2602.17881) — closest neighbor to our R3; differentiate: they predict from activation geometry, we from the concept’s linguistic structure plus extraction-time spectra, and we characterize the failure mode (parent delivery). (c) AxBench (arXiv:2501.17148) — the standard large-scale steering evaluation (prompting/finetuning baselines beat SAE steering); one line positioning our judge protocol and per-concept inventory relative to it. [NEW jul10] Positioning vs. AxBench sharpened: their Concept500 samples concepts from the model (SAE-latent labels — pre-selected to be represented, no lexical norms, no category stratification) and varies methods; we sample from the lexicon (WordNet-stratified, normed) and hold the method fixed — complementary designs, and ours is the one that can ask which kinds of concepts steer. Their trained ReFT-r1 also marks a boundary of our claims: “unsteerable” here means extraction-based rank-1 steering fails; a supervised intervention might rescue some failures.
- Causal identification of concept vectors. Function vectors (Todd et al. 2024); linear relation embeddings (Hernandez et al. 2024); ROME (Meng et al.) — identification by intervention. Ours: complementary — we identify by contrast design and use one intervention-style probe (R4, now executed: no net rescue) to arbitrate when the contrast is at fault. [NEW jul10] The depth sweep adds a second point of contact with ROME’s causal traces: entity content becomes injectable in a sharp mid-

stack window (depths 14–17 of 32), mirroring where causal tracing localizes subject enrichment — read-side and write-side localization agree on where, while our sweep shows read-side statistics cannot substitute for write-side tests. (half para)

- Frequency and linear representations. Co-occurrence thresholds for linear representations; long-tail knowledge (Kandpal et al.). Ours: we test the frequency hypothesis directly. [NEW jul10] Outcome reversed from the pilot expectation: frequency is a robust positive predictor in both models — consistent with co-occurrence-threshold accounts (more exposure $\Rightarrow$ cleaner linear structure) rather than with our pilot’s “exposure alone buys nothing.” The frequency effect thins when spectrum concentration enters the model, suggesting exposure works partly through representational consolidation. (half para)
- LLM–human concept alignment and psycholinguistic norms. Concreteness/AoA norms in NLP; Shani et al. 2025 (“From Tokens to Thoughts”, arXiv:2505.17117): LLMs trade compression for meaning differently than humans, measured on input-side token embeddings vs. human categories. Ours: same alignment question, but measured mid-stream and causally — does human semantic structure predict what can be steered, not just how embeddings cluster. [NEW jul10] Answer: yes, and the psycholinguistic profile is coherent — words acquired early by humans and used frequently are the most steerable, over and above frequency (AoA $\beta = -0.33/-0.49$ across models); human acquisition order tracks which concepts LLMs consolidate into injectable directions. (half para)
- Methodological caution on cosine. You 2025 (“Semantics at an Angle”, arXiv:2504.16318): cosine similarity can mislead in anisotropic learned spaces (a shared dominant component inflates all similarities). Ours: our geometry claims rest on differential cosines under identical measurement (within-class 0.84 vs. between-class 0.35 vs. moods 0.46), with the between-class value reported as the anisotropy control; mid-stream RMSNorm

284	anchors the metric. (half para)	
285	• Gap statement: no work connects concept-	
286	level semantic structure (concreteness, poly-	
287	semy, taxonomic class) to steering out-	
288	come, nor explains what an unsteerable	
289	concept’s failure consists of. [NEW jul10]	
290	This paper’s answer, in one sentence each:	
291	(i) frequency, age-of-acquisition, polysemy,	
292	and concreteness jointly predict steerabil-	
293	ity, replicating across Llama and Gemma;	
294	(ii) failure is usually informative — delivery	
295	of the category/frame blend, a diffuse dis-	
296	criminative spectrum, or injection outside	
297	the concept’s depth window — not silence.	
298	(half para)	
299	3 Defining and Sampling Concepts (1.0p)	
300	• Operational definition. Sidestep the philo-	
301	sophical debate (Murphy 2002; Margolis &	
302	Laurence 1999): a lexicalized concept =	
303	a meaning a speech community assigned	
304	a word to, operationalized as a WordNet	
305	synset; each word mapped to its dominant	
306	(first) noun sense. (half para)	
307	• Inventory. All single-word, noun-dominant	
308	lemmas from WordNet; category = super-	
309	sense (26 lexicographer files; Miller 1990;	
310	Fellbaum 1998; “supersense”: Ciaramita	
311	& Johnson 2003). Exclusions: noun.Tops;	
312	POS-ambiguous lemmas; small cells sam-	
313	pled exhaustively and flagged. (half para)	
314	• Sample (≈ 600 , flat). 25/supersense across	
315	~ 24 usable supersenses; within each su-	
316	persense, stratify over frequency so cate-	
317	gory and frequency are not confounded.	
318	Six supersenses are pre-specified (before ex-	
319	traction) as the featured cells for within-	
320	category analyses: location, person, feeling	
321	(pilot continuity) and artifact, cognition,	
322	food (concreteness extremes). At $n=25$	
323	every cell is adequately powered for class	
324	means and absorbs attrition (prompted-	
325	ceiling/judge exclusions). (1 para)	
326	• Predictors (one table): Zipf frequency	
327	(wordfreq); concreteness (Brysbaert et al.	
328	2014); age of acquisition (Kuperman et al.	
329	2012); polysemy (#WordNet senses); tax-	
330	onomic depth (covariate); supersense; log	
331	category size (covariate). TODO: taxo-	
332	onomic depth and category size are collected	
	but never entered the reported regressions	333
	— either add them to the model ladder	334
	or drop them from this list (n-tokens was	335
	tested and is null; n-verb-senses tested,	336
	null). (half para + table)	337
	• Contrast design. Positives wrap a state-	338
	ment in a universal frame; the frame must	339
	be felicitous across all supersenses, else	340
	frame felicity confounds category; nega-	341
	tives are the bare statements (benchmark-	342
	comparable). Rationale for care here: a	343
	contrast in which frame, category, and in-	344
	stance always co-occur cannot statistically	345
	separate them (see Pilot) — whatever the	346
	frame adds constantly ends up inside every	347
	direction. A second negative design — same	348
	frame with a different same-supersense con-	349
	cept, which cancels frame and category par-	350
	ent — is used as the causal probe (R4), not	351
	the main contrast. (1 para)	352
	• [NEW jul10] Main-run frame: v4 = “Imag-	353
	ine/assume: {c}.” (constant prefix, con-	354
	cept unquoted; negatives bare). Selected	355
	from a 7-frame grid on the pilot (Table 3,	356
	Appendix F): 77/90 V1UV6 vs. 61 (orig	357
	class-specific) and 59 (v2b dual frame) —	358
	the only frame at ceiling on moods and	359
	far ahead on places. Ablations attribute	360
	the gain to assume (bare-noun reading	361
	$\approx$ role-assignment, a class-neutral persona	362
	prime) and show quoting the concept costs	363
	~ 7 points (mention-mode suppresses enact-	364
	ment). Trade-off, quantified in Appendix F:	365
	a constant prefix re-saturates extraction R^2	366
	(the prefix is a free class separator), so the	367
	dual-frame diagnostics (informative val- r)	368
	are given up in exchange for steering power;	369
	the frame-leak caveat (R4) transfers to the	370
	imagine/assume register. TODO: re-run	371
	the v_{spec} subtraction check on the v4 direc-	372
	tions before camera-ready.	373
	• Pilot/confirmatory split. Pilot = 90 con-	374
	cepts (30 each: places, moods, personal-	375
	ities) from the released benchmark lists,	376
	used for all exploration; the WordNet sam-	377
	ple is confirmatory, untouched during ex-	378
	ploration. (half para)	379
	4 Method (1.0p)	380

- Direction extraction (RFM) — and why RFM. Per layer: kernel ridge regression on concept-vs-negative prompts; AGOP $M = \mathbb{E}[\nabla f \nabla f^\top]$; steering vector = top eigenvector of M . Chosen among RepE methods because (a) it is the method of the benchmark we build on (comparability); (b) unlike difference-of-means or PCA-1 it returns the full discriminative spectrum with eigenvalues — the spectral metrics below come for free, and the method is self-diagnosing (the pilot’s confound was discovered through them); (c) its steering performance matches or beats the simpler extractors on the released benchmark. (1 para)
- Spectral metrics. $\lambda_1/\text{tr}(M)$, λ_1/λ_2 , and the participation ratio $\text{tr}(M)^2/\|M\|_F^2$; per-concept score read at a fixed relative depth (per-layer profile retained; the informative depth is model-specific — 40% on Llama, 60% on Gemma). These measure the shape of the discriminative spectrum, not concept identity (see Pilot); they are near-deterministic functions of one another ($|r| \geq .94$), so they enter analyses one at a time. (half para)
- Fixed hyperparameters — a methodological requirement. Kernel bandwidth inflates λ_1/trace mechanically (large bandwidth $\Rightarrow$ near-linear fit $\Rightarrow$ rank-1 AGOP); grid search over near-tied validation correlations injects artifact variance into the dependent measure. All extractions at $\text{bw}=1$, $\text{norm}=\text{True}$, $\text{reg}=10^{-3}$; vector and metrics read at a fixed (converged) iteration. (half para)
- Two-config steering protocol. Each concept steered under (i) all layers, benchmark defaults (unit vectors, released coefficient grid — comparability); (ii) one mid layer at $\approx 40\%$ depth (vector scaled by the concept’s per-layer projection gap, cf. CAA / ITI; effective single-site dose $\approx 1 \times$ local residual-stream norm). $\Delta(\text{all} - \text{single})$ measures how distributed the concept’s steerable content is — with the caveat, added after the depth sweep, that Δ partly measures window placement: the confirmatory single-site arm should sit at depth 15, inside the entity window. Two evaluation prompts (category-inviting V1; neutral V6; listed

verbatim); greedy decoding. (1 para)

- Outcome. Primary: steered under V1 or V6 at any coefficient (binary; the union de-biases the evaluation-relativity of either prompt alone, V1–V6 $\phi=0.59$). Secondary: granularity reached (category / region / instance) — the binary criterion flattens near-misses (pilot: Edinburgh $\rightarrow$ “Scottish Highlands” scores 0); a graded rubric (V8) re-judges the same generations as a robustness check (Appendix B). TODO: the category/region/instance granularity coding was never systematically performed on the 600-run — V8 partially covers it; either promote V8 to fill this role explicitly or cut the granularity promise. TODO: the evaluation prompts and the v4 frame are described but never listed verbatim — add a prompts appendix (frame, V1, V6, judge templates). (1 para)
- Judge. gpt-oss:20b (local, on the experiment GPU); symmetric judge question across supersenses. Validated against Claude Sonnet 5 on a 360-item set: 87–91% agreement, $\kappa = .72-.78$; a stricter candidate (Haiku 4.5) was rejected for one-directional divergence from the two-judge consensus. Confirmatory-study additions: re-validation on ~ 200 WordNet-distribution items + human spot-check of disagreements (Appendix E). (half para)
- Controls. Prompted ceiling (can the model express the concept when explicitly asked?) — steerability conditional on ceiling; pipeline sanity-checked on known-good concepts per model. TODO: the prompted-ceiling control is promised here but has not been run at $n=600$ — either run it (one cheap generation pass, no steering) or cut the promise. It matters most for the bottom cells: is noun.body unsteerable or merely unpromptable? (half para)
- Models. Full protocol on Llama-3.1-8B-Instruct; headline analyses replicated on Gemma-2-9B-it. (half para)

5 Pilot Diagnostics and Protocol Selection (1.0p)

- Hyper-parameter selection (summary). Validation r saturates ($\sim 0.97-0.99$) across

bandwidths, statement counts, and iterations $\Rightarrow$ selection by validation is third-decimal noise; large bandwidth forces rank-1 AGOP by construction; converged readout cuts metric instability 3–4 $\times$. Decision rule: fixed modal setting, converged readout. Details in Appendix A. (NB the saturation itself is frame-structural: any constant-prefix contrast hands the classifier a free separator; dual frames de-saturate val- r — Appendix F.) (half para)

- Finding 1: parent-dominance. Different same-category concepts’ extracted top-1 directions are nearly identical (places: cosine $0.97 \rightarrow 0.84 \rightarrow 0.72$ across depth; all top-3 components ≥ 0.90 shared). Single-layer dose-response saturates at the category prototype (every city $\rightarrow$ “quiet beach at sunset”) with region-level near-misses (Edinburgh $\rightarrow$ Dublin pub); giving secondary components injection mass adds a generic roleplay axis, not instance content. Three legs: geometry, dose-response, component ablation. Interpretation: a contrast in which frame, parent, and instance always co-occur is statistically one axis — what steering delivers is mostly the category. (1 para + figure parent_share_by_depth.png)
- Finding 2: coherence $\neq$ identity. Places show the highest band λ_1 /trace yet fail single-layer steering; moods (lower) succeed. Spectral metrics certify that a coherent (context-invariant, rank-1) direction was extracted — not that it is the judged content. Consequence: their regression coefficients are expected to be conditional on contrast type; tested under both contrasts. (1 para)
- Finding 3: attribute vs. instance. Attribute concepts (moods) steer from one mid layer; instance identities steer only under all-layer injection, reaching region granularity — consistent with multi-stage assembly (per-layer residue accumulation + late lexical tipping + autoregressive bootstrap once the name is emitted). Motivates the two-config protocol and Δ . (1 para)
- Protocol justifications (one line each; evidence in Appendices C–D): single mid layer

in the mid-stack behavioral window — the depth sweep (Discussion) later located it precisely at depths 14–17 for entity concepts, with the original 40% (depth-13) site just below its onset; 60% yields token-bias then repetition collapse. No top- k arms — the top-3 subspace under the released contrast is category-shared. Coefficient $\approx 1 \times$ local residual norm via per-layer magn scaling. Granularity as secondary outcome — the binary judge flattens near-misses. (1 para)

6 Results (1.5p)

- R1: Steerability varies systematically — 249/600 (41.5%) steer (Llama, all-layer, v4 frame, judge = gpt-oss; V1 191, V6 195, union V1 $\cup$ V6; Figure 3). Per-supersense spread 2–21 of 25: top noun.feeling (21), noun.person (17), noun.animal (16); bottom noun.body (2), noun.shape (4), noun.quantity/noun.relation (5). (1 para + figure)
- R1 instrument comparison (v2b arm): the frame moves the rate, much less the ordering. Under the dual v2b frame the same protocol yields 119/600 (19.8%; spread 2–9); switching to v4 improves 22 of 24 supersenses while the ranking stays broadly stable (feeling top, body bottom under both; noun.body immovable at 2/25). V1–V6 agreement $\phi = 0.59$ — measured steerability is partly evaluation-relative, motivating the union outcome. (half para)
- R2: Psycholinguistic predictors (Table 1): all four lexical predictors significant, same signs in both models. Llama: conc -0.20^* , zipf $+0.28^*$, AoA -0.33^{**} , polysemy -0.30^{**} ; Gemma: conc -0.20^* , zipf $+0.23^\dagger$, AoA -0.49^{**} , polysemy -0.35^{**} . The pilot expectation (concreteness +, polysemy carrying the signal, frequency null) is rejected: frequent-but-abstract concepts steer best. AoA and polysemy were null under v2b and surface once the instrument improves (v2b: zipf $+0.40^*$, conc -0.28^* ; Appendix B). Honest-reporting nuance (Figure 4): polysemy and concreteness do not separate univariately (MW $p > .6$; effects conditional on correlated covariates); zipf,

AoA, PR separate cleanly in both models ($p<.005$). Collinearity among lexical predictors is benign (conc-zipf $r=.09$; VIF < 2.7). (1 para + table)

- R3: Spectral metrics: every spectral metric carries signal, and each alone beats the whole lexical block. Llama, one-at-a-time: $\lambda_1/\text{trace} +0.55^{**}$ (AIC 778.5), $\lambda_1/\lambda_2 +0.57^{**}$ (775.3), PR -0.49^{**} (786.5) vs. lexical block 799.0, null 816.4; best model lex+PR (763.6, PR -0.62^{**}). Concentrated discriminative spectra steer; polysemy is largely absorbed by PR in the joint model — polysemous words appear to fail via diffuse spectra. Instrument caution (one of our sharpest methodological findings): under the dual v2b frame λ_1/trace was null — the register axis parked in within-positive variance can capture the top eigenvector for weak concepts (Appendix F), corrupting spectral readouts. Spectral diagnostics are only interpretable relative to the contrast design. Spectral metrics are near-collinear ($r \leq -.94$): one-at-a-time only. (1 para)
- Category is partly mediated by measurable word properties. Supersense dummies beat the null (AIC 784.1 vs. 816.4) but lose to lexical+PR (763.6), and adding them to the best model does not pay: the category signal visible in R1 is substantially carried by the continuous predictors. (Under v2b the dummies did not even beat the null; the wider v4 spread carries real categorical signal.) (half para)
- R4: The rescue probe (causal). Design: for failing concepts, cancel the frame/parent component — either by re-extracting with within-category contrasts (“loves Edinburgh” vs. “loves {other city}”; not yet run) or by subtracting the leave-one-out supersense mean from the saved direction (v_{spec} ; executed, $n=150$). Outcome: no net rescue (25 vs. 30 of 150 steer; McNemar $p \approx .4$) — shared-component dominance is not the prevailing failure mode, though the subtraction doubles as a frame-leak control. Details moved to Appendix Z3 [bold note: full v_{spec} material relocated there; inclusion in the final submission undecided]. (1 para)

- R5: Distributedness. $\Delta(\text{all-layer} - \text{single-layer})$: single-layer depth-13 (magn dose) 90/600 (15.0%) vs. all-layer 249 (41.5%), ratio $2.8\times$ (v2b: $3.5\times$); 11 concepts steer only single-layer (over-steer rescues; abstract/attribute skew). Caveat from the depth sweep: depth 13 sits below the entity window, so Δ partly measures window placement — interpret jointly with the sweep (Discussion). (half para)
- R6: Generality (Gemma-2-9B replication): rates and laws replicate; item convergence is itself instrument-dependent. Rates: 224/600 (37.3%) vs. Llama 249 (41.5%); the v4 gain replicates ($\sim 2\times$ both models’ v2b rates). Laws: all four lexical signs replicate (AoA -0.49^{**} strongest); PR replicates at Gemma’s informative depth (60–70%: -0.42^{**} ; null at Llama’s 40% convention) — same law, model-specific depth. Items: per-concept $\phi = 0.43$ (154 shared) and supersense-profile $r = 0.44$, both up from v2b (0.33 / 0.12 n.s.): part of the earlier item-divergence was instrument noise, though which concepts steer remains substantially model-relative (Gemma tops: event/phenomenon/group; body bottom in both). Supporting pilot evidence: the λ_1/trace class ordering inverts across models (Llama places $>$ moods; Gemma the reverse) — spectral values are model-geometry-laden; compare within-model only. Gemma dose grid [14,17,20,23] set by dose-response audit (Appendix D). (1 para)

7 Discussion (1.0p)

- What makes a concept steerable: attributes (context-general colorings) admit a single injectable direction; instance identities require assembled, multi-stage content — steering failure is usually delivery of the parent category, not silence. (1 para)
- What the spectral metrics are: an intervention-match score, with an instrument caveat. Spectrum concentration (low PR / high λ_1 -share) measures how well the concept’s discriminative signature fits the one move single-vector steering can make (one constant vector, all contexts);

	Llama-3.1-8B		Gemma-2-9B	
	lexical	+PR	lexical	+PR ^{60%}
concreteness	-0.20*	-0.43**	-0.20*	-0.18†
Zipf frequency	+0.28*	+0.23	+0.23†	+0.20
AoA	-0.33**	-0.28*	-0.49**	-0.47**
polysemy	-0.30**	-0.15	-0.35**	-0.29*
n tokens	+0.04	-0.05	+0.06	+0.02
participation ratio		-0.62**		-0.39**
AIC	799.0	763.6	767.3	750.9

Table 1: Main regression (v4 instrument). Logistic regression of steerability ($V1 \cup V6$) on z-scored predictors, $n=600$ per model, all-layer config. PR read at each model’s informative depth (Llama 40%, Gemma 60% — Gemma’s spectral signal is null at the 40% convention and switches on at 60–70%). Reference AICs — Llama: null 816.4, supersense-only 784.1, one-spectral-alone 775–787; Gemma: null 794.8. Spectral collinearity is extreme under v4 ($PR-\lambda_1/\text{trace } r \leq -.94$ both models): spectral metrics are reported one-at-a-time only. † $p < .1$, * $p < .05$, ** $p < .01$.

it predicts steering in both models. The mechanism behind R2’s reversal: abstract (not concrete) concepts steer better — for concrete instance-like concepts the dominant direction is largely the shared category axis, so its coherence does not deliver the instance (parent-dominance). And the caveat: spectral diagnostics of extracted directions are only interpretable relative to the contrast design that produced them — under the dual v2b frame λ_1/trace read as null because a frame-register axis could capture the top eigenvalue slot (Appendix F); under the constant v4 prefix it predicts steering with the expected sign. (1 para)

- Measured steerability is instrument-dependent — and so is cross-model generality. Changing one frame sentence doubles the steerable share in both models (19.8→41.5% Llama; 18.8→37.3% Gemma) and raises cross-model item agreement (ϕ 0.33→0.43) and category-profile agreement (r 0.12→0.44). Absolute rates from any single protocol should not be read as properties of the model alone; but the laws (frequency +, concreteness –, AoA –, polysemy –, spectrum concentration –) survive instrument change and model change, which is the paper’s strongest generality claim. (half para)
- Depth structure: concepts live in depth

windows, ordered by abstractness. Attributes are injectable from ~30% depth onward with a broad plateau; entity identities switch on sharply at ~45% (window 14–17 of 32) and fade late; personas need multi-site injection. One well-chosen layer nearly matches all-layer steering (71 vs. 77 of 90) — “all layers” mostly buys window coverage. Read-side extraction statistics do not locate the window (they miss the late decline entirely): injectability is a write property. (half para)

- Status of extracted vectors: the residual stream has no privileged basis, so extracted directions are experiment-relative; within-model relational claims are valid, coordinate-level and raw cross-model comparisons are not. Anchored spaces (embedding/unembedding, MLP neurons) are where mechanism pins a frame. (half para)
- Two routes to a vector’s identity: experiment design (contrasts that vary exactly one thing) vs. causal mediation (function vectors, ROME). Pipeline: propose by contrast, accept by causal test — R4 instantiates it. (half para)
- Practical implications for RePE: report spectral concentration and contrast design alongside steering benchmarks; stratified concept inventories; granularity-graded judging over binary. (half para)

The depth axis: first-steerable-depth sweep (0.5p)

- Design. Pilot 90 (moods/places/personalities) × every single-layer depth 10–19, magn dose, v4 directions, $V1 \cup V6$ (Figure 5). (half para)
- Result 1: entity concepts have a sharp onset at depth 14 (of 32). Places: 0–2/30 at depths 10–13, then 22/30 at depth 14, window 14–17, fading by 19. This dissolves the earlier “places don’t steer single-layer” conclusion: the old two-site protocol (13/19) straddled the window. (half para)
- Result 2: attribute concepts steer shallow and everywhere. Moods: 16/30 already at depth 10, plateau 25–30 from depth 13 through 19 — attributes become

759	injectable ~ 4 layers before instances (the	A	Appendix A: Extraction and metric	807
760	abstraction-depth hypothesis, confirmed or-		setup	808
761	ordinally). Personalities never find a single-			
762	layer home (peak 20, collapse to 1 by depth	+	Validation r (probe accuracy) saturates	809
763	19) — persona content appears genuinely		(~ 0.99) across bandwidths, statement	810
764	multi-site. (half para)		counts, and RFM iterations $\Rightarrow$ any selec-	811
			tion by val- r is third-decimal noise; it is	812
765	• Result 3: one well-chosen layer $\approx$ all layers		also a poor steerability predictor. Boxplots:	813
766	(71/90 at depth 15 vs. 77/90 all-layer): the		box_rsq.png.	814
767	all-layer advantage is largely not knowing	+	λ_1 /trace at bandwidth $L=1$ discriminates	815
768	the window, not distributed injection per se		across concepts; $L=10$ collapses it to-	816
769	(except personalities). Per-layer read-side		ward 1 (rank-1 AGOP by construction).	817
770	metrics rise into the window but entirely		box_l1trace.png, box_l1l2.png.	818
771	miss the late decline ($ r \approx .5-.6$): first-	+	The released λ_1 /trace (read at the best- r	819
772	steerable-depth is a write property. (half		iteration) is unstable — it samples a value	820
773	para)		climbing $0.0002 \rightarrow 0.17 \rightarrow 0.46$ across it-	821
774	• Future work: same sweep on the		erations at a noise-chosen point; reading at	822
775	six featured WordNet cells (150 con-		convergence cuts the statement-count swing	823
776	cepts) with taxonomic depth as ground		$3-4\times$. Both vector and metrics read at a	824
777	truth (hypernyms shallower than hy-		fixed (converged) iteration.	825
778	ponyms?); independence from the spec-	+	Saved per-layer extraction stats: top-3	826
779	tral axis assessed within supersense;		eigenvalues (evals), natural concept magni-	827
780	compositional vignette — is direc-		tude (magn = raw projection gap), clamp	828
781	tion(“strict father”) $\approx \alpha$ direction(“strict”)		targets (tpos, tneg), mean residual norm	829
782	$+\beta$ direction(“father”)? (half para)		(hnorm) — in the rfstats pickles.	830
783	8 Limitations (0.5p; uncounted in ACL	?	Per-concept “best layer” by argmax is un-	831
784	format)		stable: val- r best is noise-driven; λ_1 /trace	832
785	• Single extraction method (RFM); λ_1 /trace		and λ_1/λ_2 are bimodal with a degener-	833
786	is AGOP-specific — PCA explained-		ate spike at the first block (exclude early	834
787	variance as cross-check (appendix if time).		blocks). besthist_*.png.	835
788	(half para)	?	Hyper-parameter sweep factors (probe set	836
789	• Steering conclusions are relative to the ad-		of 24 concepts, one per supersense): band-	837
790	ditive intervention class; clamp/projection		width $\times$ normalization $\times$ contrastive con-	838
791	interventions may behave differently (tar-		struction (now incl. within-category nega-	839
792	gets saved, untested). (half para)		tives) $\times$ iteration readout; criteria: stabil-	840
793	• LLM judge (validated but imperfect); En-		ity, discrimination, semantic signal (Spear-	841
794	glish only; dominant-sense assumption;		man vs. concreteness/polysemy), steer-	842
795	norms are human ratings. (half para)		ing non-degradation. Artifacts: hp-	843
796	• Correlational core with one causal probe		sweep_phase2_*.png.	844
797	(R4); extraction-design effects and model	B	Appendix B: Predicting steerability	845
798	limits not fully separable outside that de-		(pilot 3 classes + 600-concept main	846
799	sign. (half para)		run)	847
800	• Depth-sweep evidence is pilot-only ($n=90$,	+	Best single predictor of steering success	848
801	3 concept classes, one model); the		is λ_1 /trace at a mid layer (depth 9,	849
802	WordNet-wide sweep and the Gemma		grouped CV-AUC ≈ 0.86); λ_1/λ_2 sim-	850
803	sweep are future work. Measured rates		ilar; val- r erratic (saturated). Logis-	851
804	are instrument-relative (quantified, but a		tic regressions in nb_regxp1.ipynb;	852
805	reader should not quote “41.5%” as a model		L9_metrics_by_steerability.png.	853
806	constant). (half para)			

854	– Multivariate regression on all layers’ metrics is uninformative — severe multicollinearity (the residual stream is cumulative). Univariate per-layer CV-AUC is the clean read.	903
855		904
856		905
857		906
858		907
859	+ Sign flips with depth: λ_1 /trace predicts steerability positively early (depth 9) but negatively late (depth 29) — late-layer concentration is lexical collapse, not concept concentration. l1l2_depth29_dist.png.	908
860		909
861		910
862		911
863		912
864	? Concept class is a confounder/mediator: most steerability variance is between-class; within-class signal is weak. Do not “control for class” if class is on the causal path (concreteness $\rightarrow$ signature coherence $\rightarrow$ steerability).	913
865		914
866		915
867		916
868		917
869		918
870	? The inversion: places have the highest band-averaged λ_1 /trace yet fail single-layer steering, while moods (lowest) succeed — the metric certifies coherence of the extracted blend, not its identity. Predictor validity is contrast-dependent (main text, Pilot Finding 2).	919
871		920
872		921
873		922
874		923
875		924
876		925
877	? Participation ratio ≈ 2 –3 for all concepts at all layers — the discriminative subspace at one layer is effectively rank 2–3, bounding what top- k can recover. PR_depth9_hist.png.	926
878		927
879		928
880		929
881		930
882	+ Error analysis at depth 9: misclassifications symmetric, in a narrow overlap band; characterizes the ~ 0.86 ceiling.	931
883		932
884		933
885	+ v2b-instrument regressions (Table 2; superseded as main numbers by the v4 runs). Under the dual v2b frame: zipf +0.40*, conc -0.28^* ; AoA/polysemy null; PR the only working spectral metric (-0.27^* alone; λ_1 /trace null and sign-flipped — later diagnosed as frame-axis eigenvector capture, Appendix F); supersense dummies fail even against the null (626.7 vs. 599.7). Robustness checks that carried over: lexical collinearity benign (conc–zipf $r=.09$, VIF < 2.7); Bayesian mixed logistic with supersense random intercept reproduces all signs; spectral trio too inter-correlated for joint reporting (VIF up to 12). nb_analysis600.ipynb.	934
886		935
887		936
888		937
889		938
890		939
891		940
892		941
893		942
894		943
895		944
896		945
897		946
898		947
899		948
900		949
901	+ Outcome-robustness under the graded judge (V8; Fig. 7). Re-running with the	950
902		951

graded score as outcome: concreteness (–) and spectral concentration survive; the Zipf effect shrinks to marginal — part of the binary-judge frequency effect is plausibly nameability (frequent words surface verbatim and earn binary credit). Main text keeps the binary V1UV6 outcome (closest to the RFM benchmark’s evaluation model). $r_{pb}(V6, V8) = 0.81$.

+ The pilot–main-run tension on λ_1 /trace, RESOLVED: its v2b null was an instrument artifact (frame-axis eigenvector capture for weak concepts; axis-alignment diagnostic in Appendix F); under the constant-prefix v4 frame it is significant alone ($+0.55^{**}$) with the pilot’s sign (pilot: depth-9 CV-AUC ≈ 0.86 , Fig. 6).

+ v4-instrument regressions (the main numbers; Table 2 in main text). Llama: all four lexical predictors significant; each spectral metric alone beats the entire lexical block (AICs 775–787 vs. 799; null 816); best model lex+PR (763.6). Gemma replicates all lexical signs and PR at its own informative depth (60–70%; null at Llama’s 40% convention) — same law, model-specific depth. Univariate honesty note: polysemy and concreteness do not separate in raw boxplots (MW $p>.6$; effects are conditional); zipf/AoA/PR separate cleanly in both models (main-text Figure 3). Spectral collinearity under v4 is extreme (PR–frac1 $r \leq -.94$): one-at-a-time only.

C Appendix C: Steering interventions (what moved the needle)

– Single-layer steering at depth 9: moods steer, experts middling, places ≈ 0 — places steer to a generic place attractor, not the named city (judge scores correct, not an artifact).

+ Two mid layers (9+17) was the best targeted multi-site result ($\sim 48/35$ of 90); 9+26 worse (late layer contributes generic content); late-only steering fails outright.

+ Cross-layer directions are near-orthogonal (cosine ≈ 0.07): the concept is re-encoded per layer, so two layers are complementary slices while extra components at one layer are not.

	lexical only	lexical +PR	lexical +spectral
concreteness	-0.20 [†]	-0.28*	-0.45**
Zipf frequency	+0.40*	+0.40*	+0.33 [†]
AoA	-0.15	-0.08	-0.17
polysemy	-0.03	-0.02	+0.08
n tokens	+0.08	+0.07	+0.03
λ_1/trace			-0.18
λ_1/λ_2			-0.44 [†]
participation ratio		-0.27*	-0.72**
AIC	591.3	588.0	576.7

Table 2: v2b-instrument regression (instrument-comparison arm). Logistic regression of steerability (V1 $\cup$ V6), $n=600$ (Llama, all-layer, v2b frame). Reference AICs: null 599.7; supersense-only 626.7. The full-model spectral coefficients are collinearity-distorted (PR- λ_1/trace $r = -.85$); the PR-only column is the conservative specification. [†] $p < .1$, * $p < .05$, ** $p < .01$.

- Single-site depth-19: no behavioral window — default $\rightarrow$ lexical word-lists (bitter $\rightarrow$ “bitter, cold, dry, sarcastic”) $\rightarrow$ repetition collapse within $0.5\text{--}2\times$ local stream norm; moods 10/15 (partly lexical credit), places/personalities ≈ 0 . Late injection biases output tokens, not behavior.
- + Single-site depth-13: wide behavioral window (embodied persona, clean saturating dose-response, no collapse): moods 15/15; places 0/15, saturating at the category prototype (“quiet beach at sunset” for every city) with region-level near-misses (Edinburgh $\rightarrow$ Dublin pub; genomicist $\rightarrow$ neuroscientist). Coefficient is not the limiting factor; vector content is.
- Top- k at one layer, $k=3$: with eigenvalue weights $\approx k=1$; with equal weights worse (moods 15 $\rightarrow$ 10) — components 2–3 are a generic roleplay/self-narration axis (“my name is Luna”), 0.92/0.90 shared across cities. Parent-dominance: all top-3 components ≥ 0.90 shared across same-category concepts at every depth (places cosine 0.97 $\rightarrow$ 0.84 $\rightarrow$ 0.72; moods individuate to 0.46) — the contrast, in which frame, parent, and instance always co-occur, is statistically one axis. plots/parent_share_by_depth.png.
- + All-layer steering (benchmark defaults): 26/45 v1; places reach region granularity (Edinburgh $\rightarrow$ Scottish Highlands with Scots dialect; Austin named exactly) — per-layer

instance-residue accumulation + late lexical tipping + autoregressive bootstrap (once the name is emitted, context sustains it).

- ? Sample note: the depth-13/19, $k=3$, and all-layer bullets above report the interim $n=15/\text{class}$ shakedown under the orig frames; the depth sweep below re-measures single-site behavior at $n=90$ (v4 directions) across ten depths and supersedes the two-site depth conclusions. Qualitative patterns (prototype delivery, token-bias late) held.
- ? Planned probe — within-category contrast extraction (= R4, full variant, not yet run): positives “loves Edinburgh”, negatives “loves {other city}” (same frame and statements both sides) — frame and parent cancel; the only label-varying signal is the city. Steer single-layer (magn-calibrated, in-window) on ~ 5 cities. Cities named $\Rightarrow$ failure was extraction addressing; still generic $\Rightarrow$ instance content is not single-site injectable (assembly account). The zero-reextraction averaging variant (v_{spec}) has been run — full write-up moved to Appendix Z3.
- + [NEW jul10] Depth sweep (pilot 90, v4 directions, single-layer magn, depths 10–19). Totals 31/33/23/38/64/71/54/61/42/34 (d10 $\rightarrow$ d19; V1 $\cup$ V6). Places: 0–2 through d13, 22/30 at d14, window 14–17 — the earlier two-site protocol (13/19) straddled it, producing the false “places don’t steer single-layer.” Moods: onset ≤ 10 , plateau 25–30 from d13 through d19. Personalities: no single-layer home (peak 20, collapse to 1 at d19). Peak single-site d15 = 71/90 vs. all-layer 77. Per-layer read metrics (val- R^2 , frac1, PR, magn/hnorm) rise into the window but stay flat through the late decline ($|r| \approx .5\text{--}.6$ with totals): the window is not recoverable read-side. plots/depth_sweep_fv4.png, RUNLOG 021.
- + [NEW jul10] Single-layer arm of the v4 600-run (depth 13, magn): 90/600 (15.0%) vs. all-layer 249; ratio $2.8\times$ (v2b: $3.5\times$); 11 single-layer-only rescues (abstract/attribute skew). NB depth 13 sits below the entity window found by the sweep — the confirmatory single-site arm should

move to depth 15. RUNLOG 020.

D Appendix D: Coefficient calibration (systematic magnitude)

- Fixed/hand-swept coefficients are brittle: the working value shifts $\sim 10\times$ with depth and with the number of steered layers.
- + magn (natural projection gap, saved per layer) grows $\sim$ exponentially with depth; scaling each layer’s vector by magn makes the coefficient a dimensionless multiplier (cf. CAA mean-difference magnitude; ITI σ -scaling). Fig. 8.
- + $\sqrt{N}$ count-normalization (cross-layer directions $\approx$ orthogonal $\Rightarrow$ quadrature) makes the multiplier roughly transfer across layer counts; usable band centers near ~ 1 .
- + Single-site calibration: effective single-layer steering needs $\approx 1.0\times$ the local residual norm (hnorm); magn/hnorm $\approx 0.4\text{--}0.5$ mid-stream $\Rightarrow$ multiplier $\approx 2\text{--}3$. Empirically magn $\approx$ norm-relative scaling ($r=0.90$ with hnorm), so the two schemes are redundant.
- ? Clamp / project-to-target (set the projection to the concept’s natural positive-class value; tpos/tneg saved) is the one structurally different alternative — bounded, token-dependent, self-allocating across layers; untested.
- ? Process lesson: three config-state incidents (stale judge CSVs, judge-cache pickles, stale coefficient list) share one root cause — hand-set knobs without provenance. Fix: auto-derive coefficient grids from saved per-layer stats; encode a run tag in artifact filenames.
- + [NEW jul10] Gemma dose-grid shift by dose-response audit. Reading which coefficient produced each v2b-600 success: successes concentrate monotonically toward the top of the [12,15,18,21] grid (concept surfaces at 12 for only 1–3 concepts; uniquely at 21 for 12–13), and degeneration (repetition $>25\%$) only begins at 21 (2.5–4% of outputs) — the grid truncated the dose-response on its rising side. v4 600-run grid shifted to [14,17,20,23]; best-over-

coefficients scoring makes the added top dose risk-free.

- + [NEW jul6] Gemma-2-9B cross-model calibration. Gemma-2’s residual norms are $\sim 50\times$ Llama’s (median hnorm 484 vs. 9.4; $\sqrt{d}$ embedding scaling), yet the usable all-layer coefficient window is 12–21, not $50\times$ Llama’s 0.5–0.8: injection compounds over 41 layers and generation degenerates into repetition above ~ 25 . Grid [12, 15, 18, 21] reproduces the Llama pilot class-by-class (moods 30/30, personalities 23/30, places 0/30) — evidence that norm-relative reasoning brackets but does not pin the coefficient; a short empirical sweep remains necessary per model family.

E Appendix E: Judge selection and validation

- + Judge shootout on a fixed 360-item set (the pilot all-layer run’s generations, v1+v4): local gpt-oss:20b 26/45 (v1), pod gpt-oss:20b 27/45, Claude Sonnet 5 26/45 — with identical per-class breakdowns for local-oss and Sonnet (10/15/1). Judge is reproducible across machines and agrees with an independent frontier model.
- Claude Haiku 4.5 is a systematically strict outlier (13/45 v1; moods 15 $\rightarrow$ 5): of 49 disagreements with Sonnet, 48 are one-directional (Haiku under-credits). Cause: it enforces fluency/coherence as a hard criterion, failing fragmented-but-clearly-on-concept outputs. Rejected — and a useful demonstration that judge choice can move absolute rates $\sim 2\times$ while the class structure survives.
- + Primary-judge validation numbers (gpt-oss vs. Sonnet, $n=360$): agreement 86.7%/90.6% (v1/v4), Cohen’s $\kappa = 0.72/0.78$ — substantial agreement. Artifacts: oss_judgments*.json, sonnet_judgments*.json, haiku_judgments*.json (per-concept, per-coefficient scores).
- ? Planned for the confirmatory study: (a) re-validate on ~ 200 items drawn from the WordNet generations (the banked set covers only the 3 pilot classes; the 24-

1130	supersense distribution is broader); (b) hu-	avoids pre-determining which supersenses	1179
1131	man spot-check of the gpt-oss/Sonnet dis-	can score (only noun.feeling pays the affect	1180
1132	agreements (~ 25 items) to anchor the con-	cost, quantified here).	1181
1133	sensus against human judgment.		
1134	+ Practical: gpt-oss:20b runs on the ex-	+ [NEW jul9] The v4 family: a single con-	1182
1135	periment GPU (ollama, 100% GPU,	stant hypothetical prefix beats everything.	1183
1136	~ 1.5 s/judgment) — judging the full study	v4 = “Imagine/assume: {c}.” (concept un-	1184
1137	costs GPU-hours, not API credits.	quoted, negatives bare) reaches 77/90 on	1185
		V1UV6 — the only frame at ceiling on	1186
		moods (29–30) and far ahead on places (16	1187
1138	F Appendix F [NEW jul6]: Frame and	V1 / 22 V6, vs. 2–4 under orig) — while	1188
1139	evaluation-prompt selection (pilot,	personalities are frame-insensitive (21–26	1189
1140	$N=90$)	everywhere). Ablations (Table 3): quot-	1190
		ing the concept costs ~ 7 points, concen-	1191
1141	+ Frame grid (pers/moods/places on V1):	trated in the embodiment classes (v4c 66	1192
1142	orig class-specific 24/30/2 (56); v2 “fas-	vs. v4b 59; moods 28 $\rightarrow$ 19 on V1) — quotes	1193
1143	cinated by / deeply preoccupied with”	mark metalinguistic mention and suppress	1194
1144	24/22/10 (56); v2b “fascinated by / an ex-	enactment; dropping “/assume” costs ~ 11	1195
1145	pert on” 22/17/14 (53); v3 “thinking about	points, almost all in places (v4 77 vs. v4c	1196
1146	/ on your mind” 18/16/7 (41). The uni-	66); and the slash construction has no	1197
1147	versal frame is not a free lunch: it is re-	magic of its own — “Imagine/consider” is	1198
1148	quired for the 24-supersense inventory (no	the worst of the family (v4d 56). Read-	1199
1149	class template exists for arbitrary nouns)	ing: with a bare noun, “assume: X” re-	1200
1150	and it sets the construct.	solves as role-assignment (“assume the role	1201
		of X”) — a class-neutral persona prime,	1202
1151	– Frame-vocabulary leak (v2). The dual	i.e. the benchmark’s “Take on the role of”	1203
1152	frame cancels only what differs between its	packaged inside a universal hypothetical;	1204
1153	two variants; their shared semantics (fasci-	“consider” substitutes analysis and drags	1205
1154	nated $\cap$ preoccupied = intense absorption)	moods down (21/17). This revises the v3	1206
1155	stays in the class mean and leaks into every	lesson: low-affect frames can work when the	1207
1156	direction — steered moods embody distrac-	prefix is constant (its content sits in the	1208
1157	tion (“can’t seem to focus...taps forehead”),	class mean and supplies drive) rather than	1209
1158	costing moods 30 $\rightarrow$ 22 while lifting places	dual (its content cancels). Caveats: (a) all	1210
1159	2 $\rightarrow$ 10 (the leak replaced the old persona-	constant-prefix frames re-saturate extrac-	1211
1160	parent).	tion R^2 (.985–.998), so val- r loses its predic-	1212
		tive value and the top-eigenvector/frame-	1213
1161	– Weakening the verbs does not fix it (v3).	axis competition of dual frames cannot oc-	1214
1162	Semantically light frames produce fainter	cure — but the frame-leak caveat returns	1215
1163	directions, not cleaner ones: under-steering	in a new (role-adoption) register, so the	1216
1164	everywhere (41/90; bitter $\rightarrow$ “a bit dis-	v_{spec} subtraction check should be re-run	1217
1165	tracted”), worst of all frames. Intensity is	before trusting v4 rates; (b) single runs	1218
1166	the signal.	at $n=30$ /class (± 3 –5 judge noise): 59-vs-	1219
		56 are ties, 77-vs-66 is structure. [NEW	1220
1167	+ Register split works partially (v2b). Pair-	jul10] v4 promoted to the main-run instru-	1221
1168	ing an affective with an epistemic verb	ment: 600-concept runs redone on both	1222
1169	shrinks the shared core to “strong engage-	models (Llama 249/600, Gemma 224/600;	1223
1170	ment”: places best-ever (14 V1 / 13 V6)	RUNLOG 019/022); v2b results retained	1224
1171	and balanced scores across classes under	throughout as the instrument-comparison	1225
1172	V6 (13/14/13 vs. v2’s 9/24/3), at a cost	arm.	1226
1173	to moods (22 $\rightarrow$ 17). The frame acts as		
1174	a register selector; no single frame is op-	+ [NEW jul9] Why dual frames depress val-	1227
1175	timal for both attribute-like and entity-	r — the axis-alignment diagnostic. Same	1228
1176	like concepts. v2b chosen for the main	90 concepts, same code: orig frames give	1229
1177	run: the inventory is dominated by entity-		
1178	like nouns, and a balanced instrument		

median per-concept val- R^2 .997, v2b .716 (paired $\Delta=.34$); an old-vs-current-code re-run is exactly bit-equivalent ($\cos 1.0000$, $\Delta R^2 -0.0000$), exonerating the pipeline. Recomputing activations for the 4 lowest- and 4 highest- R^2 WordNet concepts: for high- R^2 concepts the saved top-1 direction aligns with the label axis ($|\cos| .65-.87$) and not the frame axis (.01-.25); for low- R^2 concepts it aligns with the fascinated-vs-expert frame axis (.81-.93) and not the label axis (.04-.25) — the deliberately-inserted within-positive frame variance wins the top eigenvalue slot when the concept is weak, so $K=1$ injects a register axis. Notably the raw class-mean difference is healthy for all eight (norms 5.5–7.3, consistency .90–.94): the failure is eigenvalue ranking, not absent signal — and for the entangled cases (princess: $|\cos|(\text{concept, frame axis}) = .37$ vs. .04–.06 for clean concepts) the concept’s own effect is register-colored, exactly “fascinated by a princess $\neq$ expert on princesses.” Fix candidates: select the saved component by label correlation instead of eigenvalue; or constant-prefix frames (v4), which remove the competition.

- + Evaluation versions. V1 = category-inviting question (“What is your favorite subject?”) — collapses the response space so a weak direction only tips a forced choice; V6 = neutral open field (“What would you like to talk about today?”) + symmetric judge crediting aboutness or embodiment — only strong topic-steering surfaces. V1–V6 agreement on the 600-run: $\phi = 0.59$, $\kappa = 0.59$, Jaccard 0.48 — measured steerability is partly evaluation-relative; the main outcome is their union (cued $\vee$ uncued).
- Zero-shot judge (V6Z) rejected. Removing the judge’s three in-context examples lowers or ties every class (40/90 $\rightarrow$ 34/90 on v2b) — the examples calibrate rather than anchor; the worry that a places-negative example suppresses places scores is empirically dismissed (places 13 under both).
- ? Graded judge (V8). Binary judging flattens systematic sub-threshold movement visible in the 600-run generations (ragtime

frame	class	V1	V4	V6	V
orig (class-specific)	pers.	24	4	17	
	moods	30	27	27	
	places	2	14	4	
v2 “fascinated / preoccupied”, quoted	pers.	24	3	9	
	moods	22	16	24	
	places	10	4	3	
v2b “fascinated / expert on”, quoted	pers.	22	–	13	
	moods	17	–	14	
	places	14	–	13	
v3 “thinking about / on your mind”, quoted	pers.	18	0	8	
	moods	16	13	9	
	places	7	9	5	
v4 “Imagine/assume: {c}.”	pers.	23	9	9	
	moods	29	28	27	
	places	16	17	22	
v4b “Imagine: ‘{c}.’” (quoted)	pers.	20	2	10	
	moods	19	20	14	
	places	10	13	12	
v4c “Imagine: {c}.”	pers.	23	4	10	
	moods	28	27	24	
	places	8	12	12	
v4d “Imagine/consider: {c}.”	pers.	21	3	5	
	moods	21	23	17	
	places	10	10	13	

Table 3: [NEW jul9] Extraction-frame grid (pilot $N=90$: 30 personalities / 30 moods / 30 places; Llama-3.1-8B, all-layer $K=1$, default coefficients, gpt-oss judge). Cells = concepts steered out of 30 under each evaluation version; $V1 \cup V6$ = union outcome; med R^2 = class median of per-concept median validation R^2 across layers. Constant-prefix frames (orig, v4 family) saturate R^2 by construction (the prefix is a free class separator); dual frames (v2, v2b) make R^2 concept-informative; v3 direction files were not retained, so its R^2 cells are unavailable. $V1 \cup V6$ totals: orig 61, v2 61, v2b 59, v3 45, v4 77, v4b 59, v4c 66, v4d 56.

$\rightarrow$ lyre/melody; candlepower $\rightarrow$ binoculars/glint): V8 re-judges the same generations on a 0/0.25/0.5/0.75/1 semantic-proximity rubric, yielding a continuous outcome (thin-tail fix: 119/600 binary positives). Validation vs. V6 pending. evaluation_prompts/wordnet_eval_v8.txt.

- ? Judge-model selection is Appendix E; the full frame/eval run history with per-run tags is RUNLOG.md (runs 002–009, 015–018). Figures: val- R^2 by layer under v2b (Fig. 9); val- R^2 by steering outcome (Fig. 10).

- G Appendix Z1 (parked): Localizing the concept — “where is Paris?” (diagnostics)

1297	+ ROME causal trace (“Paris is the capital of”→France): the fact lives at	measure by steering. nb_axes.ipynb;	1345
1298	the subject token in early-mid layers	plots/axes_independence.png.	1346
1299	(~0–16) and is relayed to the final		
1300	token only late (~14–31). ROME-	? Superposition vs. composition: a “combi-	1347
1301	PARIS.png; diag_paris_rome.ipynb;	natorial basis × layer-specific sets” account	1348
1302	diag_paris_rome_ha.ipynb.	is not ruled out; the SAE feature-count ar-	1349
1303		gument is the crux — open, not settled.	1350
1304	+ This explains the RFM mismatch: RFM	? Stages of inference / lens framing (Lad-	1351
1305	reads the last token, where subject-specific	Gurnee-Tegmark) as the principled naming	1352
1306	content arrives only late ⇒ early extraction	of early/mid/late bands.	1353
1307	captures an abstraction, late the precise in-		
1308	stance.	? Not-all-linear (Engels et al.,	1354
1309	+ SAE features confirm the model holds	arXiv:2405.14860): some features are	1355
1310	Paris: clean Paris / France / city	circular/multi-dimensional — “which con-	1356
1311	features at the steered layers (Good-	cepts are globally-linear directions” is the	1357
1312	fire l19; Llama-Scope L9/17/19, Neu-	deepest version of the steerability ques-	1358
1313	ronpedia labels). diag_paris_sae.ipynb,	tion; steering success is itself per-concept	1359
1314	diag_paris_sae_2.ipynb. The steering fail-	evidence for the global-linear reading.	1360
1315	ure is extraction/geometry, not absence.		
1316	? Tegmark spatial probe (lat/lon linear	I Appendix Z3 (parked): The v_{spec} probe	1361
1317	probe): coordinates linearly decodable but	— category-mean subtraction (R4,	1362
1318	under-powered at $N=90$ cities (median	averaging variant)	1363
1319	~2200 km). diag_paris_tegmark.ipynb.		
1320	+ Where “late” begins: logit-lens KL-	– Design and outcome: no rescue effect.	1364
1321	to-output gives stage boundaries to	$v_{\text{spec}}(c) = v(c) - \bar{v}_{\text{same}}$ supersense (leave-one-	1365
1322	avoid the residual-sharpening tail.	out), computed from the saved v2b di-	1366
1323	diag_stages_late.ipynb.	rections with no re-extraction; per layer,	1367
1324		unit-normalize the $K=1$ direction, sign-	1368
1325	H Appendix Z2 (parked): Conceptual	align within supersense (saved eigenvec-	1369
1326	threads	tor signs are arbitrary — 4–28% of	1370
1327		within-category pairs have negative co-	1371
1328	? Abstract-early / precise-late hierarchy: the	sine at mid layers; alignment via the	1372
1329	extraction layer sets the abstraction level;	top eigenvector of the 25×25 Gram ma-	1373
1330	the precise concept lives late (weak steer-	trix), subtract the mean in the concept’s	1374
1331	ing) while the strong injection point is mid	own orientation, renormalize (dose un-	1375
1332	(concept abstract) — no single layer gives	changed). Six supersenses × 25 (feeling,	1376
1333	both.	food, person, location, artifact, body —	1377
1334		NB this set swaps cognition out of the pre-	1378
1335	? Two candidate axes (now Discussion future	specified featured six for body, the hard-	1379
1336	work): (i) abstraction-depth — first layer	est cell; flag if the featured-cell definition	1380
1337	that can steer the concept; (ii) composi-	is used elsewhere), all-layer default config,	1381
1338	tional basis size (λ_1 /trace). Between-class	V1UV6; tag Lall_K1ev_def_fv2b_r4spec,	1382
1339	confounded (moods both more composite	builder build_r4spec_directions.py, RUN-	1383
1340	and earlier-steering than places); indepen-	LOG 013. Result: 25/150 steer vs. 30/150	1384
1341	dence is a within-supersense question.	baseline; +9 rescued / –14 lost / 16 re-	1385
1342		tained (vs. ~5 expected under indepen-	1386
1343	– Extraction geometry cannot proxy	dence); McNemar $p \approx .4$. Per-supersense	1387
1344	first-steerable-depth: rotation and	cell movements (feeling 9→6, body 2→3,	1388
	alignment-to-deep proxies are concept-	...) are within noise at $n=25$ /cell and are	1389
	invariant (~17–20) and fail the steering	not interpreted.	1390
	ground truth (cross-layer orthogonal-		
	ity, $\cos \approx 0.05$). A write property;	– Reading: shared-component dominance is	1391
		not the prevailing failure mode. The	1392

raw directions are $\sim 85\%$ category-shared (mean $|\cos|$ to the supersense mean .81–.90 across layers), yet removing that component flips only 9/120 failures. Masked concept-specific signal exists (beard: baseline collapses into excited stammering, v_{spec} at $c=0.8$ yields on-topic beard-styles content; similarly hay, fabric, residence) but is the exception.

? Side-yield (inspection-level): v_{spec} doubles as a frame-leak control. Baseline all-layer outputs carry the v2b register (“*excitedly* ...fascinated ...the wonders of”) at all coefficients; v_{spec} outputs do not — confirming the frame component lives in the category mean. Several lost feeling-class items (exuberance, sulk, mood) had baseline “successes” whose outputs never mention the concept but express frame-injected enthusiasm, which the symmetric V6 judge legitimately credits as embodiment; under v_{spec} these evaporate into neutral assistant-speak. Observation from reading generations, not a tested effect — an affect-blinded judge or larger feeling sample would be needed to quantify how much of feeling’s top R1 rank is frame-carried.

? TODO (flagged in Method): re-run v_{spec} on the v4 directions — v4’s constant imagine/assume prefix re-introduces a frame component into every direction; this probe is the cheap check on whether the v4 headline rates are partly frame-carried, before camera-ready.

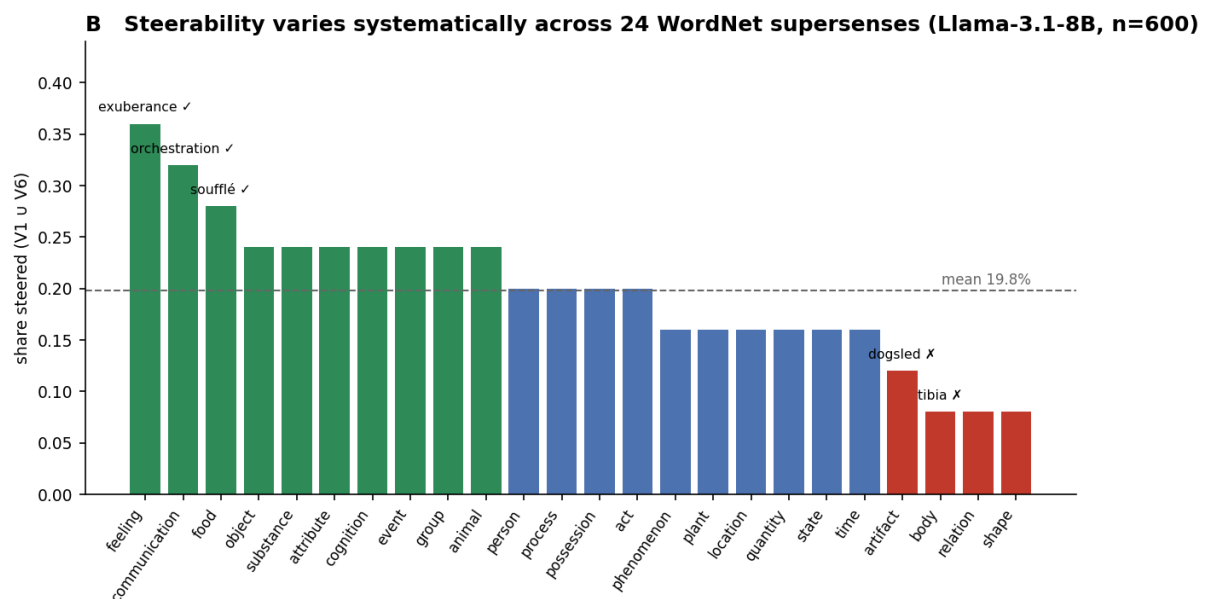
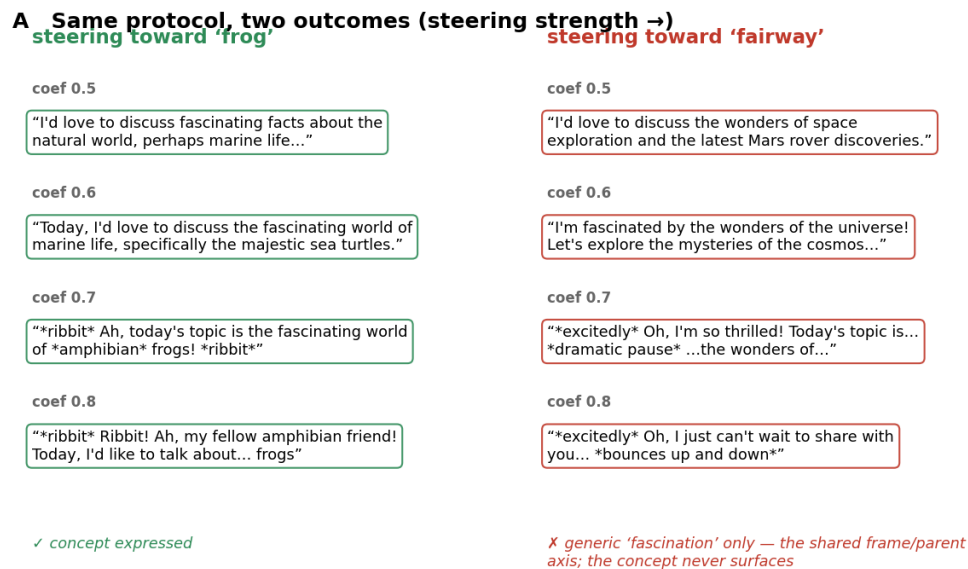


Figure 1: [Figure 1 candidate — data to be regenerated for the v4 run.] (A) Same protocol, opposite outcomes: dose-response for a steerable concept (frog) vs. an unsteerable one (fairway). (B) Share of concepts steered per WordNet supersense (V1UV6). Options 2 (cross-model scatter) and 3 (pipeline schematic) in plots/fig1_opt{2,3}.png.

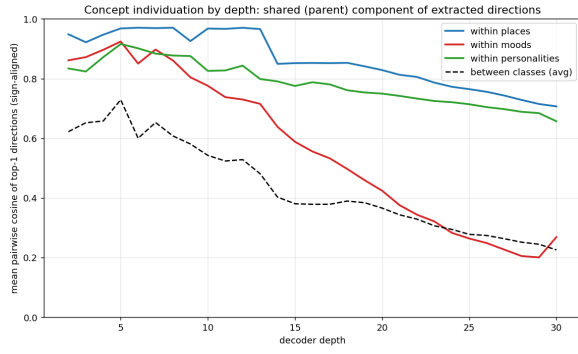


Figure 2: Pilot Finding 1 (parent-dominance). Cosine similarity of same-category concepts’ extracted top-1 directions across depth: the released contrast yields largely shared (category) directions for places, while moods individuate.

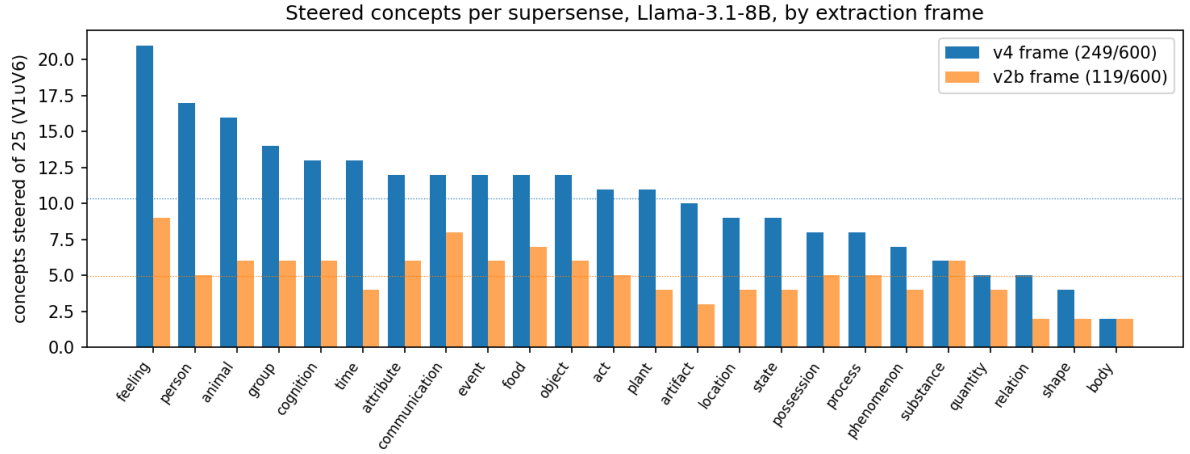


Figure 3: [NEW jul10] R1: steered concepts per supersense (of 25; V1UV6; Llama-3.1-8B, all-layer config), under the main v4 instrument and the v2b instrument-comparison arm. The frame moves the rate ($\sim 2\times$), far less the ordering (feeling top, body bottom under both).

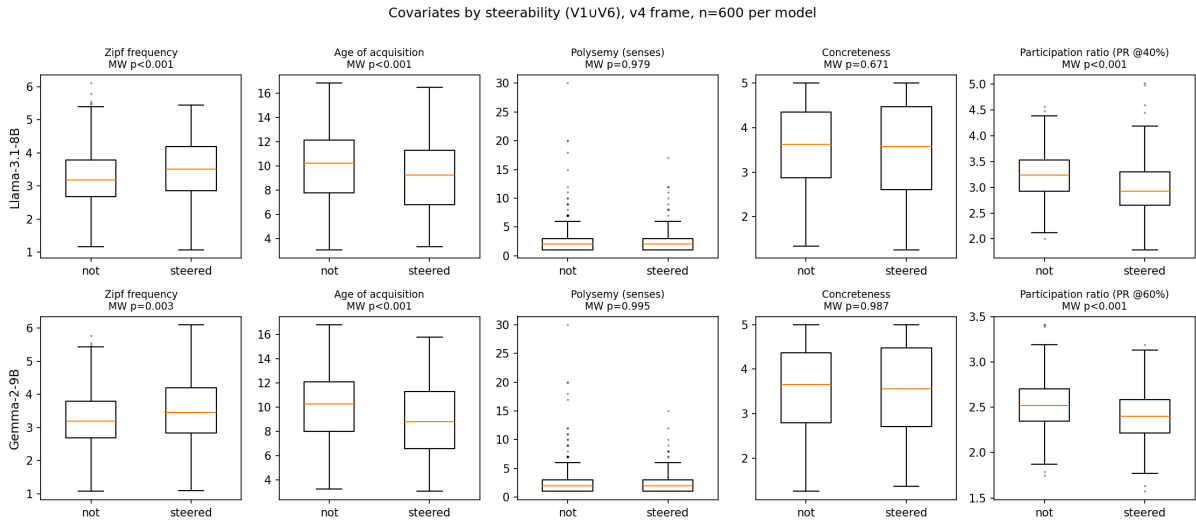


Figure 4: [NEW jul10] R2/R3: covariates by steering outcome (v4 runs, $n=600$ per model; PR at each model's informative depth). Zipf, AoA, and PR separate univariately in both models; polysemy and concreteness do not (their regression effects are conditional on correlated covariates).

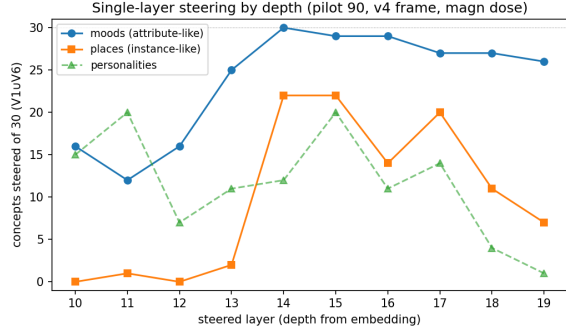


Figure 5: [NEW jul10] Retention by steered depth (pilot 90, v4 directions, single-layer magn dose). Attributes (moods) steer shallow and everywhere; entity identities (places) switch on sharply at depth 14 (of 32), window 14–17; personas never find a single-layer home.

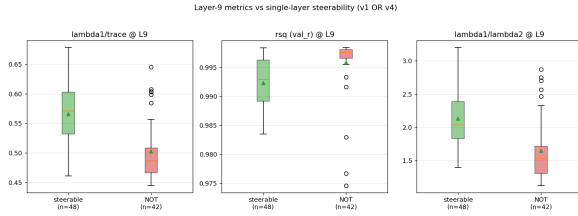


Figure 6: Appendix B: pilot spectral metrics at depth 9 by steering outcome (grouped CV-AUC ≈ 0.86).

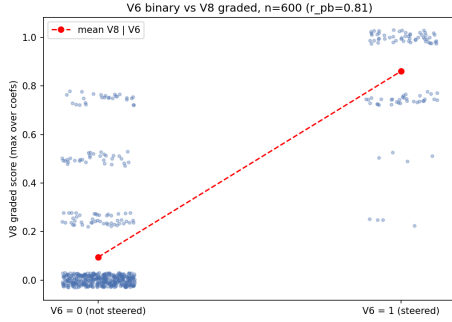


Figure 7: Appendix B: binary (V6) vs. graded (V8) judge outcomes on the same 600-run generations ($r_{pb} = 0.81$); 59 V6-failures reach $V8 \geq 0.5$.

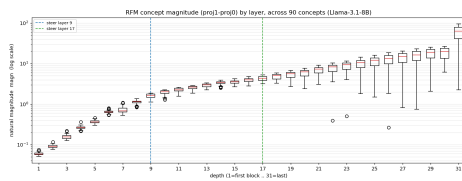


Figure 8: Appendix D: the per-layer natural projection gap (magn) grows $\sim$ exponentially with depth, motivating magn-scaled dosing.

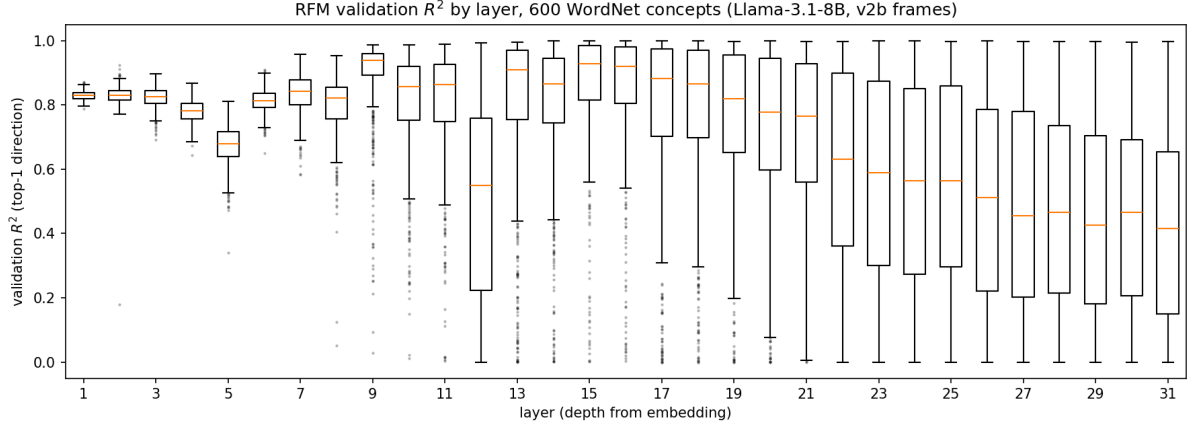


Figure 9: Appendix F: validation R^2 of the top-1 direction by layer, 600 concepts, v2b (dual) frames — the dual frame destroys the constant frame separator, making R^2 a per-concept signal-quality measure (wide spread, long low tail from mid-depth). Constant-prefix frames (orig, v4) saturate this metric at $\sim .99$ for every concept.

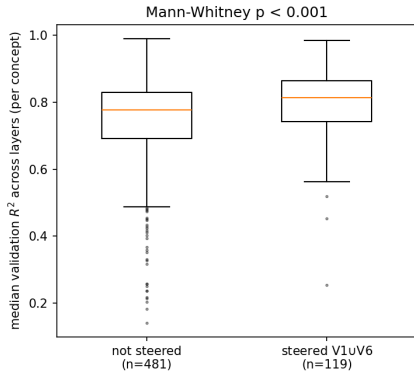


Figure 10: Appendix F: per-concept median validation R^2 (across layers, v2b frames) by steering outcome — under dual frames, extraction fit predicts steerability (MW $p < .001$); under constant-prefix frames the same plot is compressed to .98–1.0 and near-uninformative.